

Pharmaceutical Reform and Pharmacy Entry

County-level Results, v2: One-Year Effective Policy Timing

Wenxia Chen

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Empirical Setting

- ▶ Unit of analysis: county-by-year panel.
- ▶ Controlled sample: 2012–2019, with population, nighttime lights, and road density.
- ▶ v2 timing treats the reform as effective in the year after the policy year:

$$Reform_{ct} = \mathbf{1}\{t \geq PolicyYear_c + 1\}.$$

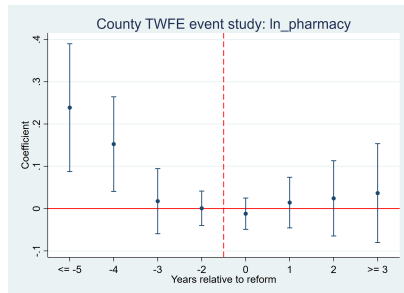
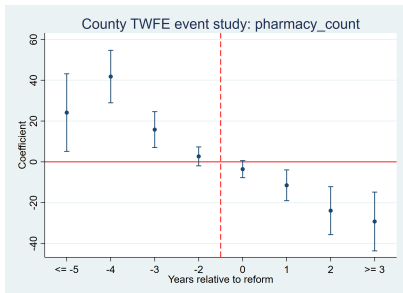
- ▶ Standard errors are clustered at the county level.
- ▶ TWFE and CSDID event-study figures use the same outcome definitions and event-time window as the five-estimator robustness figures.

TWFE Baseline Results: Count and ln(count)

	Count	ln(count)
Post policy	3.739* (1.565)	0.031* (0.014)
County FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
N	17,463	15,940
R ²	0.842	0.876

Controls include log population, log nighttime lights, and log road density.

TWFE Event-study Results



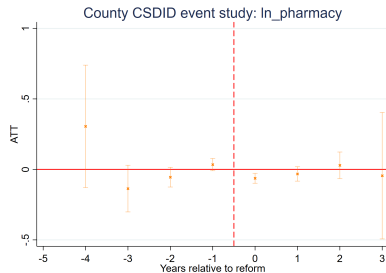
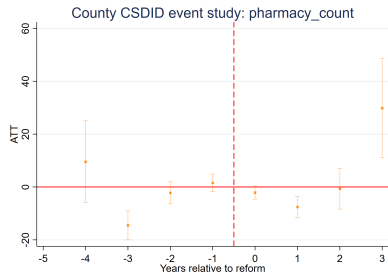
Relative year -1 is omitted. County and year fixed effects are absorbed.

CSDID Results

	Count	ln(count)
ATT	-2.608 (1.819)	-0.040 (0.026)
Estimator	CSDID	CSDID
Controls	Yes	Yes
Control group	Not-yet-treated	Not-yet-treated

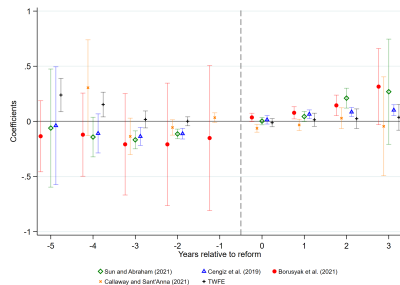
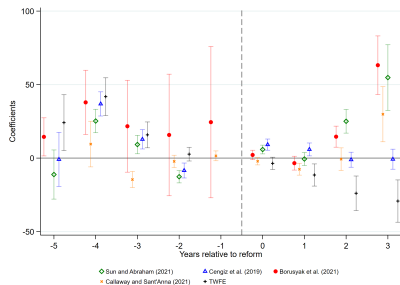
CSDID uses Callaway–Sant’Anna group-time ATT with doubly robust inverse probability weighting and county-clustered standard errors.

CSDID Event-study Results



Same event window as the TWFE and five-estimator event-study figures; relative year -1 is omitted.

Five-estimator Event-study Robustness



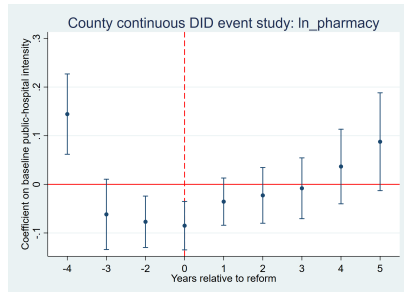
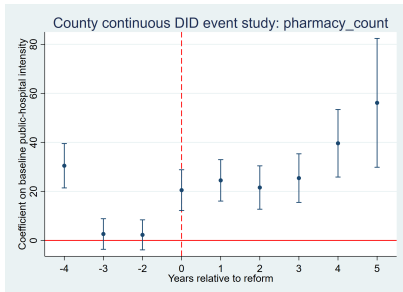
Estimators: TWFE, Callaway–Sant’Anna CSDID, Sun–Abraham interaction-weighted, Borusyak–Jaravel–Spiess imputation, and stacked event study.

Continuous DID: OLS and IV Baseline Results

	OLS		IV: 1959 stock	
	Count	ln(count)	Count	ln(count)
Post $\times$ ln(public hospitals)	20.308*** (3.460)	-0.028 (0.021)	79.687*** (13.734)	0.107 (0.083)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	17,463	15,940	14,241	12,886
R ² / K-P F	0.843	0.876	100.124	147.517

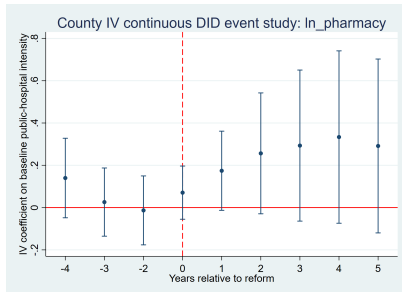
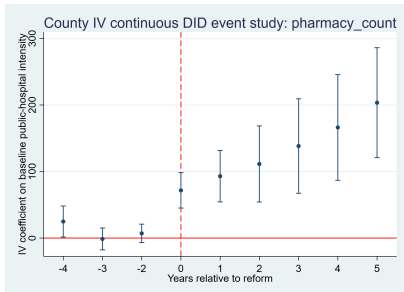
IV instruments interact post policy with the 1959 epidemiology-station stock.

Continuous DID Event-study: OLS



Event-time indicators are interacted with baseline public-hospital intensity.

Continuous DID Event-study: IV



Dynamic IV first stages are weak, so these figures are diagnostic rather than the main IV evidence.

Takeaways

- ▶ TWFE shows smaller positive count and log-count effects after retaining 2012 policy-year counties.
- ▶ CSDID static ATT remains negative for count and log count, so staggered-timing robustness does not support the TWFE increase.
- ▶ Five-estimator event-study figures place TWFE and CSDID in the same event-time comparison set.
- ▶ Continuous DID OLS remains positive for pharmacy counts; static IV estimates are larger and have strong first stages.
- ▶ IV event-study plots are retained as diagnostics; the main IV evidence is the static continuous-DID specification.